

40. Write the inverse in words and determine its truth value.

2.6 De Morgan's Laws



Figure 2.12 De Morgan's Laws were key to the rise of logical mathematical expression and helped serve as a bridge for the invention of the computer. (credit: modification of work "Golden Gate Bridge (San Francisco Bay, California, USA)" by James St. John/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Use De Morgan's Laws to negate conjunctions and disjunctions.
2. Construct the negation of a conditional statement.
3. Use truth tables to evaluate De Morgan's Laws.

The contributions to logic made by Augustus De Morgan and George Boole during the 19th century acted as a bridge to the development of computers, which may be the greatest invention of the 20th century. **Boolean** logic is the basis for computer science and digital electronics, and without it the technological revolution of the late 20th and early 21st centuries—including the creation of computer chips, microprocessors, and the Internet—would not have been possible. Every modern computer language uses Boolean logic statements, which are translated into commands understood by the underlying electronic circuits enabling computers to operate. But how did this logic get its name?



PEOPLE IN MATHEMATICS

George Boole

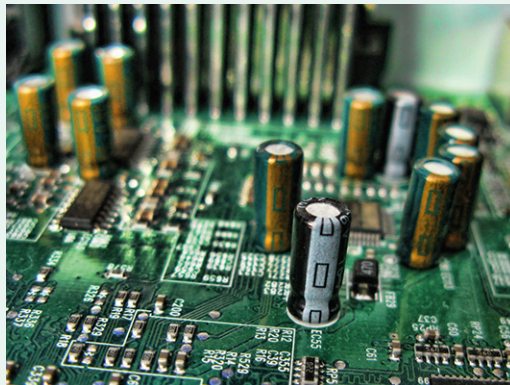


Figure 2.13 Boole's algebra of logic was foundational in the design of digital computer circuits. (credit: "Circuit Board" by Squeazyboy/Flickr, CC BY 2.0)

George Boole was born in Lincolnshire, England in 1815. He was the son of a cobbler who provided him some initial education, but Boole was mostly self-taught. He began teaching at 16 years of age, and opened his own school at the age of 20. In 1849, at the age of 34, he was appointed Professor of Mathematics at Queens College in Cork, Ireland. In 1853, he published the paper, *An Investigation of the Laws of Thought, on Which Are Founded the Mathematical Theories of Logic and Probabilities*, which is the treatise that the field of Boolean algebra and digital circuitry was built on.

Reference: Posamentier, Alfred and Spreitzer Christian, "Chapter 35 George Boole: English (1815-1864)" pp. 279-283, *Math Makers: The Lives and Works of 50 Famous Mathematicians*, Prometheus Books, 2019.

Negation of Conjunctions and Disjunctions

In Chapter 1, [Example 1.37](#) used a Venn diagram to prove **De Morgan's Law** for set complement over union. Because the complement of a set is analogous to negation and union is analogous to an *or* statement, there are equivalent versions of De Morgan's Laws for logic.

FORMULA

De Morgan's Law for negation of a conjunction: $\sim(p \wedge q) \equiv \sim p \vee \sim q$

De Morgan's Law for the negation of a disjunction: $\sim(p \vee q) \equiv \sim p \wedge \sim q$

Negation of a conditional: $\sim(p \rightarrow q) \equiv p \wedge \sim q$

Writing conditional as a disjunction: $p \rightarrow q \equiv \sim p \vee q$

 Recall that the symbol for logical equivalence is: $\equiv$.

De Morgan's Laws allow us to write the negation of conjunctions and disjunctions without using the phrase, "It is not the case that ...," to indicate the parentheses. Avoiding this phrase often results in a written or verbal statement that is clearer or easier to understand.

EXAMPLE 2.26

Applying De Morgan's Law for Negation of Conjunctions and Disjunctions

Write the negation of each statement in words without using the phrase, "It is not the case that."

1. Kristin is a biomedical engineer and Thomas is a chemical engineer.
2. A person had cake or they had ice cream.

Solution

1. Kristin is a biomedical engineer and Thomas is a chemical engineer has the form " $p \wedge q$," where p is the statement, "Kristin is a biomedical engineer," and q is the statement, "Thomas is a chemical engineer." By De Morgan's Law, the negation of a conjunction, $\sim(p \wedge q)$, is logically equivalent to $\sim p \vee \sim q$. $\sim p$ is "Kristin is not a biomedical engineer," and $\sim q$ is "Thomas is not a chemical engineer." By De Morgan's Law, the solution has the form " $\sim p \vee \sim q$," so the answer is: "Kristin is not a biomedical engineer or Tom is not a chemical engineer."
2. A person had cake or they had ice cream has the form " $p \vee q$," where p is the statement, "A person had cake," and q is the statement, "A person had ice cream." By De Morgan's Law for the negations of a disjunction, $\sim(p \vee q) \equiv \sim p \wedge \sim q$. The solution is the statement: "A person did not have cake and they did not have ice cream."

YOUR TURN 2.26

Write the negation of each statement in words without using the phrase, *it is not the case that*.

1. Jackie played softball or she ran track.
2. Brandon studied for his certification exam, and he passed his exam.

Negation of a Conditional Statement

The negation of any statement has the opposite truth values of the original statement. The **negation of a conditional**, $\sim(p \rightarrow q)$, is the conjunction of p and not q , $p \wedge \sim q$. Consider the truth table below for the negation of the conditional.

p	q	$p \rightarrow q$	$\sim(p \rightarrow q)$
T	T	T	F
T	F	F	T
F	T	T	F
F	F	T	F

The only time the negation of the conditional statement is true is when p is true, and q is false. This means that $\sim(p \rightarrow q)$ is logically equivalent to $p \wedge \sim q$, as the following truth table demonstrates.

p	q	$p \rightarrow q$	$\sim(p \rightarrow q)$	$\sim q$	$p \wedge \sim q$	$\sim(p \rightarrow q) \leftrightarrow (p \wedge \sim q)$
T	T	T	F	F	F	T
T	F	F	T	T	T	T
F	T	T	F	F	F	T
F	F	T	F	T	F	T

EXAMPLE 2.27

Constructing the Negation of a Conditional Statement

Write the negation of each conditional statement.

1. If Adele won a Grammy, then she is a singer.
2. If Henrik Lundqvist played professional hockey, then he did not win the Stanley Cup.

Solution

1. The negation of the conditional statement, $p \rightarrow q$, is the statement, $p \wedge \sim q$. The hypothesis of the conditional statement is p : "Adele won a Grammy," and conclusion of the conditional statement is q : "Adele is a singer." The negation of the conclusion, $\sim q$, is the statement: "She is not a singer." Therefore, the answer is $p \wedge \sim q$: "Adele won a Grammy, and she is not a singer."
2. The hypothesis is p : "Henrik Lundqvist played professional hockey," and the conclusion of the conditional statement is q : "He did not win the Stanley Cup." The negation of q is the statement: "He won the Stanley Cup." The negation of the conditional statement is equal to $p \wedge \sim q$: "Henrick Lundqvist played professional hockey, and he won the Stanley Cup."



YOUR TURN 2.27

Write the negation of each conditional statement.

1. If Edna Mode makes a new superhero costume, then it will not include a cape.
2. If I had pancakes for breakfast, then I used maple syrup.

EXAMPLE 2.28**Constructing the Negation of a Conditional Statement with Quantifiers**

Write the negation of each conditional statement.

1. If all cats purr, then my partner's cat purrs.
2. If a penguin is a bird, then some birds do not fly.

✓ **Solution**

1. The negation of the conditional statement $p \rightarrow q$ is the statement $p \wedge \sim q$. The hypothesis of the conditional statement is p : "All cats purr," and the conclusion of the conditional statement is q : "My partner's cat purrs." The negation of the conclusion, $\sim q$, is the statement: "My partner's cat does not purr." Therefore, the answer is $p \wedge \sim q$: "All cats purr, but my partner's cat does not purr."
2. The hypothesis is p : "A penguin is a bird," and the conclusion of the conditional statement is q : "Some birds do not fly." The negation of q is the statement: "All birds fly." Therefore, the negation of the conditional statement is equal to $p \wedge \sim q$: "A penguin is a bird, and all birds fly."

> **YOUR TURN 2.28**

Write the negation of each conditional statement.

1. If some people like ice cream, then ice cream makers will make a profit.
2. If Raquel cannot play video games, then nobody will play video games.

Many of the properties that hold true for number systems and sets also hold true for logical statements. The following table summarizes some of the most useful properties for analyzing and constructing logical arguments. These properties can be verified using a truth table.

Property	Conjunction (AND)	Disjunction (OR)
Commutative	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
De Morgan's	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
Conditional	$\sim(p \rightarrow q) \equiv p \wedge \sim q$	$p \rightarrow q \equiv \sim p \vee q$

EXAMPLE 2.29**Negating a Conditional Statement with a Conjunction or Disjunction**

Write the negation of each conditional statement applying De Morgan's Law.

1. If mom needs to buy chips, then Mike had friends over and Bob was hungry.
2. If Juan had pizza or Chris had wings, then dad watched the game.

✓ **Solution**

1. The conditional has the form "If p then q or r ," where p is "Mom needs to buy chips," q is "Mike had friends over," and r is "Bob was hungry." The negation of $p \rightarrow (q \wedge r)$ is $p \wedge \sim(q \wedge r)$. Applying De Morgan's Law to the statement $\sim(q \wedge r)$ the result is $\sim q \vee \sim r$, so our conditional statement becomes $p \wedge (\sim q \vee \sim r)$. By the distributive property for conjunction over disjunction, this statement is equivalent to $(p \wedge \sim q) \vee (p \wedge \sim r)$. Translating the statement $(p \wedge \sim q) \vee (p \wedge \sim r)$ into words, the solution is: "Mom needs to buy chips and Mike did not have friends over, or Mom needs to buy chips and Bob was not hungry."

2. The conditional has the form “If p or q , then r ,” where p is “Juan had pizza,” q is “Chris had wings,” and r is “Dad watched the game.” The negation of $(p \vee q) \rightarrow r$ is $(p \vee q) \wedge \sim r$. By the distributive property for disjunction over conjunction, the statement is equivalent to $(p \vee \sim r) \wedge (q \vee \sim r)$. Translating the statement $(p \vee \sim r) \wedge (q \vee \sim r)$ into words, the solution is: “Juan had pizza or dad did not watch the game, and Chris had wings or dad did not watch the game.”

> YOUR TURN 2.29

Write the negation of each conditional statement applying De Morgan's Law.

1. If Eric needs to replace the light bulb, then Marcus left the light on all night or Dan broke the bulb.
2. If Trenton went to school and Regina went to work, then Merika cleaned the house.

Evaluating De Morgan's Laws with Truth Tables

In Chapter 1, you learned that you could prove the validity of De Morgan's Laws using Venn diagrams. Truth tables can also be used to prove that two statements are logically equivalent. If two statements are logically equivalent, you can use the form of the statement that is clearer or more persuasive when constructing a logical argument.

The next example will prove the validity of one of De Morgan's Laws using a truth table. The same procedure can be applied to any two logical statement that you believe are equivalent. If the last column of the truth table is a tautology, then the two statements are logically equivalent.

EXAMPLE 2.30

Verifying De Morgan's Law for Negation of a Conjunction

Construct a truth table to verify De Morgan's Law for the negation of a conjunction, $\sim(p \wedge q) \equiv \sim p \vee \sim q$, is valid.

✓ Solution

Step 1: To verify any logical equivalence, you must first replace the logical equivalence symbol, $\equiv$, with the biconditional symbol, $\leftrightarrow$. The statement $\sim(p \wedge q) \equiv \sim p \vee \sim q$ becomes $\sim(p \wedge q) \leftrightarrow \sim p \vee \sim q$.

Step 2: Next, you create a truth table for the statement. Because we have two basic statements, p , and q , the truth table will have four rows to account for all the possible outcomes. The columns will be p , q , $\sim p$, $\sim q$, $p \wedge q$, $\sim(p \wedge q)$, $\sim p \vee \sim q$, and the biconditional statement is $\sim(p \wedge q) \leftrightarrow \sim p \vee \sim q$.

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$\sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$
T	T	T	F	F	F	F	T
T	F	F	T	F	T	T	T
F	T	F	T	T	F	T	T
F	F	F	T	T	T	T	T

Step 3: Finally, verify that the statement is valid by confirming it is a tautology. In this instance, the last column is all true. Therefore, the statement is valid and De Morgan's Law for the negation of a conjunction is verified.

> YOUR TURN 2.30

1. Construct a truth table to verify De Morgan's Law for the negation of a disjunction, $\sim(p \vee q) \equiv \sim p \wedge \sim q$, is valid.

Check Your Understanding

30. De Morgan's Law for the negation of a conjunction states that $\sim(p \wedge q)$ is logically equivalent to _____.
31. De Morgan's Law for the negation of a disjunction states that $\sim(p \vee q)$ is logically equivalent to _____.
32. The negation of the conditional statement, $\sim(p \rightarrow q)$, is logically equivalent to _____.
33. $\sim(\sim(p \rightarrow q)) \equiv p \rightarrow q$, which means the conditional statement is logically equivalent to $\sim(p \wedge \sim q)$. Apply _____ to the statement $\sim(p \wedge \sim q)$ to show that the conditional statement $p \rightarrow q \equiv \sim p \vee q$.



SECTION 2.6 EXERCISES

For the following exercises, use De Morgan's Laws to write each statement without parentheses.

1. $\sim(\sim p \vee q)$
2. $\sim(\sim p \wedge \sim q)$
3. $\sim(p \wedge \sim q)$
4. $\sim(\sim p \vee \sim q)$

For the following exercises, use De Morgan's Laws to write the negation of each statement in words without using the phrase, "It is not the case that, ..."

5. Sergei plays right wing and Patrick plays goalie.
6. Mario is a carpenter, or he is a plumber.
7. Luigi is a plumber, or he is not a video game character.
8. Ralph Macchio was the original Karate Kid, and karate is not for defense only.
9. Some people like broccoli, but my siblings did not like broccoli.
10. Some people do not like chocolate or all people like pizza.

For the following exercises, write each statement as a conjunction or disjunction in symbolic form by applying the property for the negation of a conditional.

11. $\sim(p \rightarrow q)$
12. $\sim(p \rightarrow \sim q)$
13. $\sim(\sim p \rightarrow q)$
14. $\sim(\sim p \rightarrow \sim q)$
15. $\sim(p \wedge q \rightarrow \sim r)$
16. $\sim(p \rightarrow q \vee r)$
17. $\sim(p \vee q \rightarrow \sim r)$
18. $\sim(p \rightarrow q \wedge r)$

For the following exercises, write the negation of each conditional in words by applying the property for the negation of a conditional.

19. If a student scores an 85 on the final exam, then they will receive an A in the class.
20. If a person does not pass their road test, then they will not receive their driver's license.
21. If a student does not do their homework, then they will not play video games.
22. If a commuter misses the bus, then they will not go to work today.
23. If a racecar driver gets pulled over for speeding, then they will not make it to the track on time for the race.
24. If Rene Descartes was a philosopher, then he was not a mathematician.
25. If George Boole invented Boolean algebra and Thomas Edison invented the light bulb, then Pacman is not the best video game ever.
26. If Jonas Salk created the polio vaccine, then his child received the vaccine or his child had polio.
27. If Billie Holiday sang the blues or Cindy Lauper sang about true colors, then John Lennon was not a Beatle.
28. If Percy Jackson is the lightning thief and Artemis Fowl is a detective, then Artemis Fowl will catch Percy Jackson.
29. If all rock stars are men, then Pat Benatar is not a rock star.
30. If Lady Gaga is a rock star, then some rock stars are women.
31. If yellow combined with blue makes green, then all colors are beautiful.
32. If leopards have spots and zebras have stripes, then some animals are not monotone in color.

For the following exercises, construct a truth table to verify that the logical property is valid.

33. $p \rightarrow q \equiv \sim p \vee q$
34. $p \rightarrow \sim q \equiv \sim p \vee \sim q$
35. $\sim p \rightarrow q \equiv p \vee q$
36. $\sim(p \rightarrow \sim q \vee \sim r) \equiv p \wedge q \wedge r$